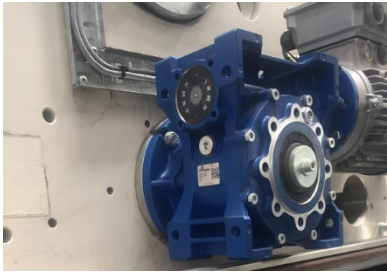


| Maintenance Requirement                                              | Equipment & Task to Be Performed                                                                                                                          |
|----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Quarterly</b>                                                     | Remove the shutter cylinder housing covers and clean any debris from the shutter guide rails, rollers and cylinders. Replace the covers before operation. |
| <p>Figure 12.2-105<br/>Shutter cylinder.</p>                         |                                                                         |
| <b>Quarterly</b>                                                     | Remove and clean the gearbox breather plug using acetone                                                                                                  |
| <p>Figure 12.2-106<br/>Location of the<br/>gearbox breather plug</p> |                                                                        |
| Maintenance Requirement                                              | Equipment & Task to Be Performed                                                                                                                          |
| <b>Annual</b>                                                        | Replace the oil in the gearbox.                                                                                                                           |
| <p>Figure 12.2-107 Oil level<br/>indicator</p>                       |                                                                       |

| Maintenance Requirement                                                     | Equipment & Task to Be Performed                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Annual</b></p>                                                        | <p>Visual inspection of the lifting cables, checking for damage or strand breaks.</p> <p>When broken wires are detected, the number and position, along with the examiner's opinion of the general rope condition, will decide whether or not the rope should be discarded.</p> <p>If local concentrations or groups of broken wires are found, the rope should be discarded when:</p> <ul style="list-style-type: none"> <li>• Three or more broken wires are found in close proximity of the termination.</li> <li>• Three or more broken wires are found in one strand</li> <li>• Five broken strands are found between two adjacent strands within a length of 10 x rope diameter.</li> </ul> |
| <p>Figure 12.2-108<br/>Example of a shield lifting cable showing damage</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Maintenance Requirement                                                     | Equipment & Task to Be Performed                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <p><b>3 Years</b></p>                                                       | <p>Under the UK Health &amp; Safety guidelines, It is recommended that the lifting cables be replaced every 3 years.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

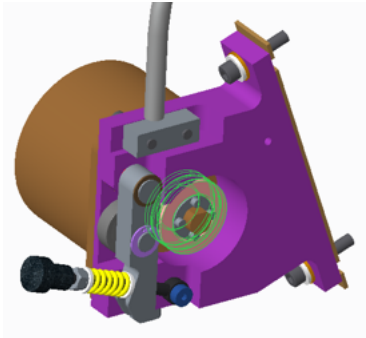
12.2.20 Aluminium Wire Feeder Units



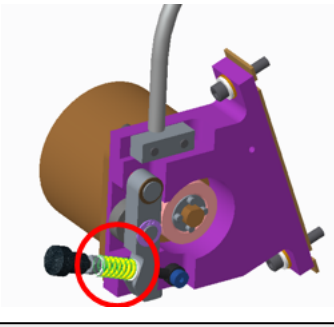
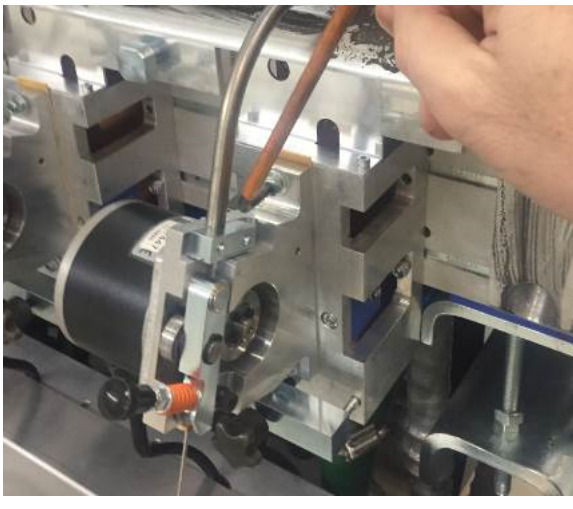
Figure 12.2-109 Wire feeder units with and without the protective cover

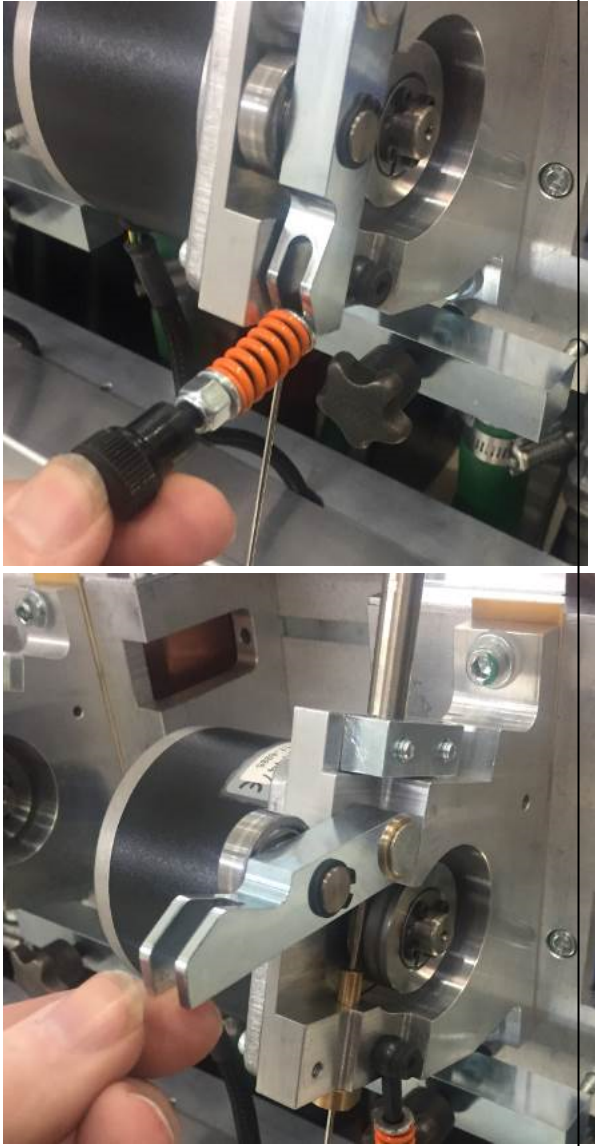
| Maintenance Requirement                                                                  | Equipment & Task to Be Performed         |
|------------------------------------------------------------------------------------------|------------------------------------------|
| Weekly                                                                                   | Check tightness of the wire feeder clamp |
| <p>Figure 12.2-110 Example of a technician checking the wire feeder guide tube clamp</p> |                                          |

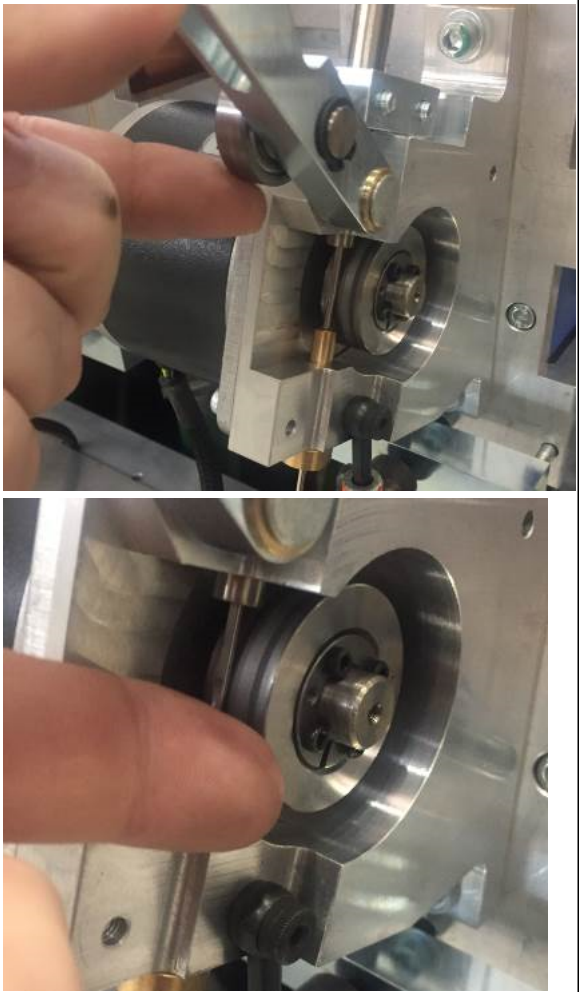
| Maintenance Requirement                                                                      | Equipment & Task to Be Performed                                                                                                                                 |
|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Weekly</b>                                                                                | Check the condition of the lower guide tube for wear. Excessive wear will create the wire to wander. This can be considered a wear part, and will need replacing |
| Figure 12.2-111 Example of a technician checking the condition of the wire feeder guide tube |                                                                               |

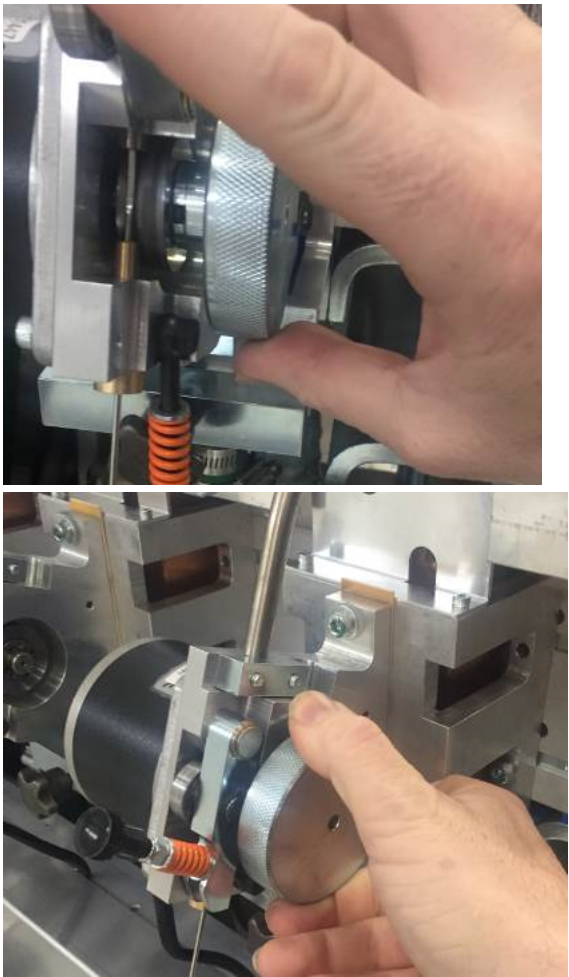
| Maintenance Requirement                                                   | Equipment & Task to Be Performed                                                     |
|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <b>Monthly</b>                                                            | Turn the drive wheel onto the second groove or clean with caustic soda               |
| Figure 12.2-112 Schematic of the wire feeder highlighting the drive wheel |  |

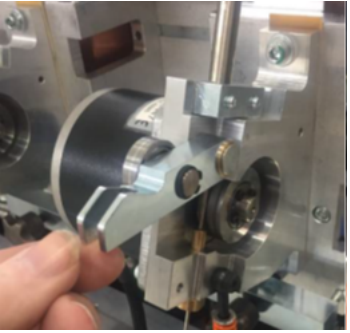
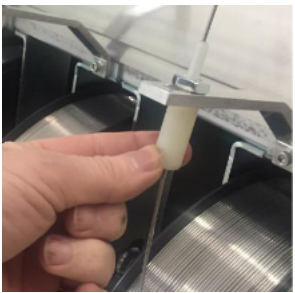


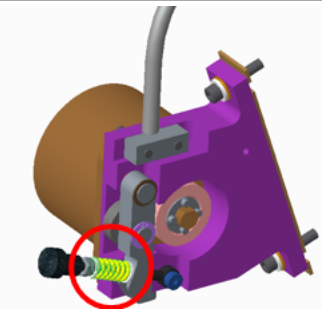
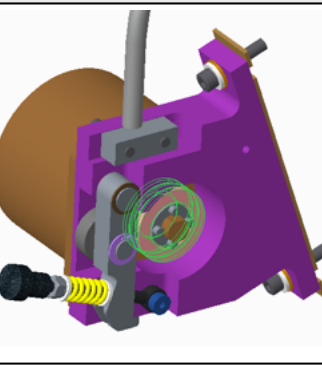
| Maintenance Requirement                                                                | Equipment & Task to Be Performed                                                     |
|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <b>Monthly</b>                                                                         | Check spring compression                                                             |
| <p>Figure 12.2-113 Schematic of the wire feeder highlighting the spring</p>            |    |
| <b>Monthly</b>                                                                         | Clean the wire feeder spouts with caustic soda                                       |
| <p>Figure 12.2-114 Schematic of the wire feeder highlighting the wire feeder spout</p> |   |
| <b>Monthly</b>                                                                         | Check the wire feeder holding fasteners are tight.                                   |
| <p>Figure 12.2-115 Example of a technician checking the fasteners</p>                  |  |

| Maintenance Requirement                                                                  | Equipment & Task to Be Performed                                                    |
|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Monthly                                                                                  | Check for correct compression of spring and ease of movement of pivot.              |
| <p data-bbox="304 813 708 875">Figure 12.2-116 Example of a technician making checks</p> |  |

| Maintenance Requirement                                                                                     | Equipment & Task to Be Performed                                                     |
|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Monthly                                                                                                     | Check rotation of sealed bearing/wheel and the surface condition of the drive wheels |
| <p>Figure 12.2-117 Check rotation of sealed bearing/wheel and the surface condition of the drive wheels</p> |   |

| Maintenance Requirement                                             | Equipment & Task to Be Performed                                                    |
|---------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Monthly                                                             | Check for smooth rotation of all components.                                        |
| <p>Figure 12.2-118 Check for smooth rotation of all components.</p> |  |

| Maintenance Requirement                                                  | Equipment & Task to Be Performed                                                      |
|--------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Quarterly                                                                | Clean the rotating arms of any dirt                                                   |
| Figure 12.2-119 Example of a technician cleaning the rotating arms       |     |
| Figure 12.2-120 Example of a technician cleaning the rotating arms       |     |
| Quarterly                                                                | Check the condition of plastic infeed bush                                            |
| Figure 12.2-121 Example of a technician checking the plastic infeed bush |  |
| Quarterly                                                                | Check the rotation of the bearing                                                     |
| Figure 12.2-122 Schematic of the wire feeder highlighting the bearing    |  |

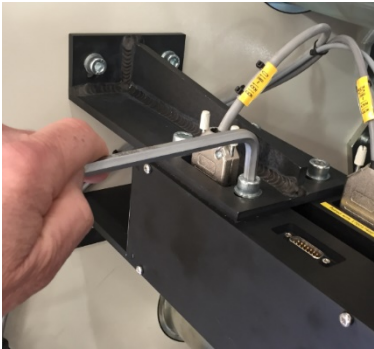
| Maintenance Requirement                                                   | Equipment & Task to Be Performed                                                                                                                                                                                                                                                                                                            |
|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bi-Annual                                                                 | Replace the bearings                                                                                                                                                                                                                                                                                                                        |
| Figure 12.2-123 Schematic of the wire feeder highlighting the bearing     |  <p>A 3D schematic of a wire feeder assembly. The main body is purple, and a large brown wheel is visible on the left. A yellow spring is located at the bottom left. A green circle highlights a bearing on the central shaft where the wire passes.</p> |
| Bi-annual                                                                 | Replace the springs                                                                                                                                                                                                                                                                                                                         |
| Figure 12.2-124 Schematic of the wire feeder highlighting the spring      |  <p>A 3D schematic of the wire feeder assembly, similar to the previous one. A red circle highlights the yellow spring located at the bottom left of the assembly.</p>                                                                                    |
| Bi-annual                                                                 | Replace the drive wheels                                                                                                                                                                                                                                                                                                                    |
| Figure 12.2-125 Schematic of the wire feeder highlighting the drive wheel |  <p>A 3D schematic of the wire feeder assembly. A green circle highlights the large brown drive wheel on the left side of the assembly.</p>                                                                                                             |



### 12.2.21 Inline Optical Monitor System

| Maintenance Requirement                                                                              | Equipment & Task to Be Performed                                                                                                                 |
|------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Weekly                                                                                               | Check all the interconnection cables for damage.                                                                                                 |
| Weekly                                                                                               | Check the monitor beam sensors are working correctly from the HMI diagnostic screen, as explained in this manual, under the calibration section. |
| Weekly                                                                                               | Wipe the beam surfaces with a lint free cloth to keep the lenses clean with a soft felt (optical grade)                                          |
| Figure 12.2-126 Example of a technician checking the cables between the faceplate and monitor system |                                                                |

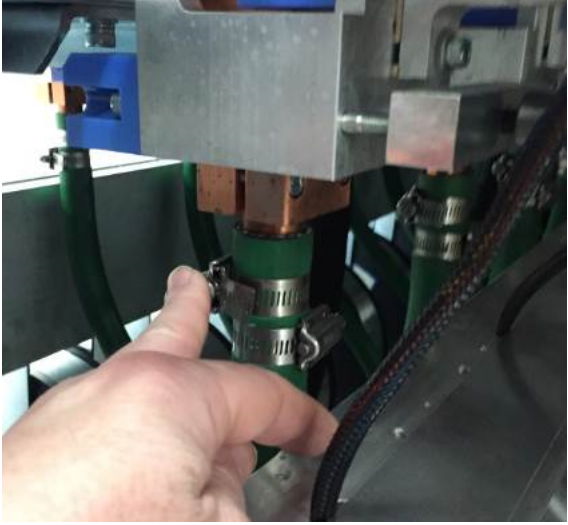
| Maintenance Requirement | Equipment & Task to Be Performed                                                                             |
|-------------------------|--------------------------------------------------------------------------------------------------------------|
| Monthly                 | Calibrate the system. <a href="#">See chapter 11.2 Optical In-Line Monitor Probe Calibration on page 526</a> |

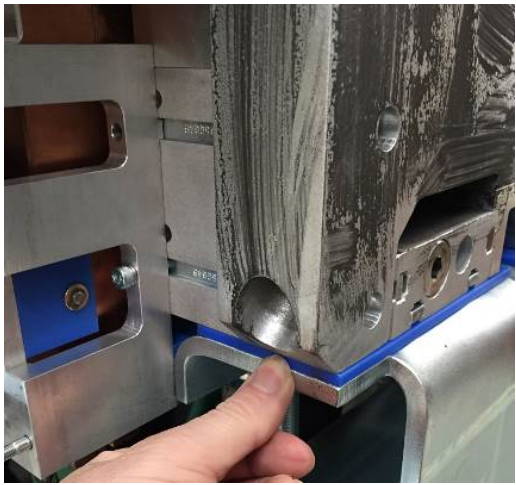
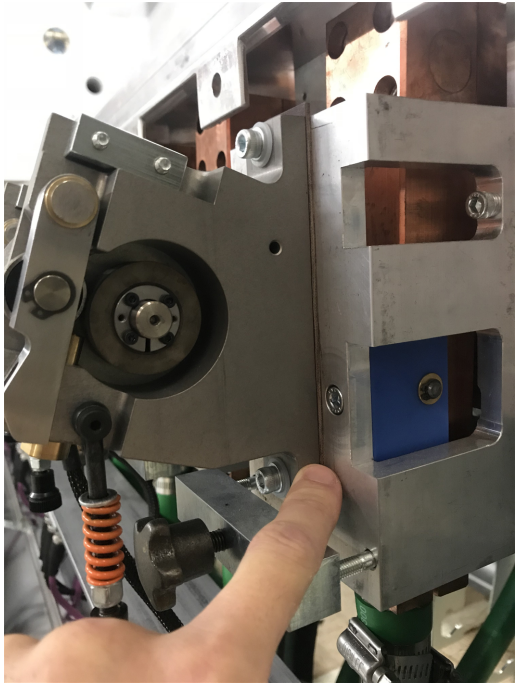
| Maintenance Requirement                                                           | Equipment & Task to Be Performed                                                     |
|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Quarterly                                                                         | Check the tightness of all the holding bolts                                         |
| Figure 12.2-127 Example of a technician checking the location bolts for tightness |  |

| Maintenance Requirement                                            | Equipment & Task to Be Performed                                                                                                                           |
|--------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Annually                                                           | Replace the filter material in the beam enclosure. This may need to be carried out more regularly, if there is large amounts of airborne dust in the area. |
| Figure 12.2-128 The location of the filter material being replaced |                                                                          |

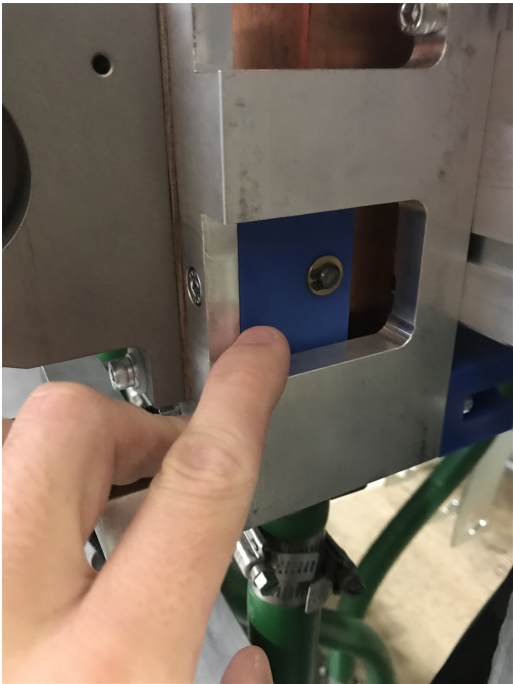
## 12.2.22 Evaporation Source Area

| Maintenance Requirement                                                                  | Equipment & Task to Be Performed                                                                                                                                                                              |
|------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Weekly</b>                                                                            | Ensure that after cleaning, all areas are given general visual inspection.                                                                                                                                    |
| <b>Weekly</b>                                                                            | Visually check the evaporation source area removable shielding for damage. Repair or replace as required.                                                                                                     |
| Figure 12.2-129 Example of the removable shields on the evaporation source area          |                                                                                                                             |
| <b>Weekly</b>                                                                            | <p>Visually check all water cooling lines and plugs for indication of leakage. Resolve as necessary.</p> <p>Enable the water-cooling from the water diagnostic HMI screen, with the cart fully retracted.</p> |
| Figure 12.2-130 Example of a technician checking a waterline for the viewing source flap |                                                                                                                           |

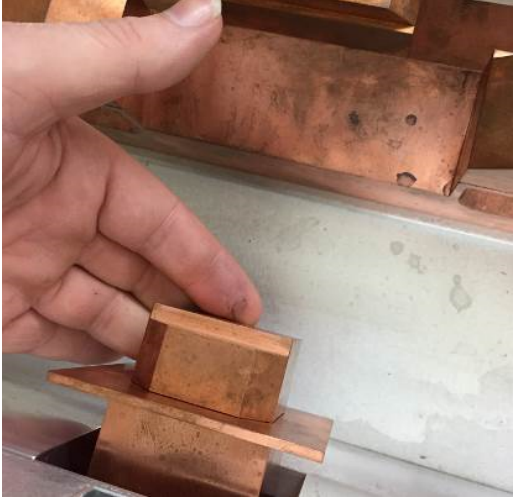
| Maintenance Requirement                                                                                          | Equipment & Task to Be Performed                                                                                                                                                                                                                                                                                                             |
|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Weekly</b>                                                                                                    | Visually check the heat resistant hose cover is in place, as it will protect the rubber hoses from radiant heat damage                                                                                                                                                                                                                       |
| <p>Figure 12.2-131 Example of a technician checking a waterline to the live post area</p>                        |  A close-up photograph showing a person's hand pointing at a green plastic waterline that is connected to a metal fitting on a piece of industrial equipment. The equipment has various other components, including a blue hose and a metal bracket.       |
| <p>Figure 12.2-132 Example of a technician checking a waterline to the evaporation source enclosure assembly</p> |  A close-up photograph showing a person's hand checking a connection between a grey waterline and a metal enclosure. The hand is holding the end of the waterline, which is inserted into a metal fitting. The enclosure is made of metal plates.         |
| <p>Figure 12.2-133 Example of a technician checking a waterlines to the evaporation source area</p>              |  A close-up photograph showing a person's hand checking a connection between a brass waterline and a metal enclosure. The hand is holding the end of the brass waterline, which is inserted into a metal fitting. The enclosure is made of metal plates. |

| Maintenance Requirement                                                                                                                                                               | Equipment & Task to Be Performed                                                                                            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Monthly                                                                                                                                                                               | Visually check the insulators around the source area and wire feeders for electrical shorts or damage. Replace as required. |
| <p>Figure 12.2-134 Example of a technician checking the evaporation base insulator to ensure that there is no indication of electrical short circuits taking place.</p>               |                                           |
| <p>Figure 12.2-135 Example of a technician checking the aluminium wire feeder-mounting insulator to ensure that there is no indication of electrical short circuits taking place.</p> |                                          |



| Maintenance Requirement                                                                                                                                                                     | Equipment & Task to Be Performed                                                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <p>Figure 12.2-136 Example of a technician checking the evaporation power live posts mounting insulator to ensure that there is no indication of electrical short circuits taking place</p> |   |
| <p>Monthly</p>                                                                                                                                                                              | <p>Check on the tightness of the copper block retaining bolts</p>                   |
| <p>Figure 12.2-137 Example of a technician checking the tightness of the copper block locating bolts</p>                                                                                    |  |



| Maintenance Requirement                                                                                                                    | Equipment & Task to Be Performed                                                                                                                                                                                                                   |
|--------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Monthly</b></p>                                                                                                                      | <p>Check the condition of the cabling and plugs feeding power to each of the stepper motors. Make repairs as necessary</p>                                                                                                                         |
| <p>Figure 12.2-138 Example of a technician checking the condition of the wiring to the stepper motors</p>                                  |                                                                                                                                                                  |
| <p><b>Monthly</b></p>                                                                                                                      | <p>Check the condition of the force that can be applied to the evaporators from the spring loaded contact blocks.</p> <p>BML have special equipment for measuring this force and can advise the customer of a supplier for the test equipment.</p> |
| <p>Figure 12.2-139 Example of a technician checking the force required to move the copper contact block</p>                                |                                                                                                                                                                 |
| <p>Figure 12.2-140 Example of a technician checking the force required to move the copper contact block using a force measuring device</p> |                                                                                                                                                                |

| Maintenance Requirement                                                                      | Equipment & Task to Be Performed                                                   |
|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Monthly                                                                                      | Check the condition of the universal joint and “rubber boot”                       |
| <p>Figure 12.2-141 Example of a technician checking the condition of the universal joint</p> |  |